1.1

Yes, because each state already contains all the information needed to predict the future evolution of the system.
There is no additional variable like how long we have been in a state, that affects the next transition
Therefore, the probability of the next state depends only on the current state, not on the sequence of past states.

1.2.

new state: S5 (Failure / Dead Battery)

Example transition matrix

$$P = \begin{bmatrix} 0.4 & 0.4 & 0.2 & 0.0 & 0.0 \\ 0.2 & 0.5 & 0.2 & 0.1 & 0.0 \\ 0.1 & 0.4 & 0.4 & 0.1 & 0.0 \\ 0.0 & 0.1 & 0.1 & 0.2 & 0.6 \\ 0.0 & 0.0 & 0.0 & 0.0 & 1.0 \end{bmatrix}$$

1. 3. Degraded States

$\left.\begin{array}{l} \text{S1D (Sunny/Full)} \\ \text{S2D (Cloudy/Low)} \\ \text{S3D (Night/Low)} \\ \text{S4D (Night/Empty)} \end{array}\right\}$ Low Charge

$$P = \begin{bmatrix} 0.4 & 0.4 & 0.2 & 0 & 0 & 0 & 0 & 0 \\ 0.2 & 0.5 & 0.2 & 0.1 & 0 & 0 & 0 & 0 \\ 0.1 & 0.4 & 0.4 & 0.1 & 0 & 0 & 0 & 0 \\ 0 & 0.3 & 0.1 & 0.2 & 0 & 0 & 0 & 0.4 \\ 0 & 0 & 0 & 0 & 0.3 & 0.5 & 0.2 & 0 \\ 0 & 0 & 0 & 0 & 0.1 & 0.6 & 0.2 & 0.1 \\ 0 & 0 & 0 & 0 & 0.05 & 0.45 & 0.4 & 0.1 \\ 0 & 0 & 0 & 0 & 0 & 0.4 & 0.2 & 0.4 \end{bmatrix}$$

2.1 $V_2 = V_0 \times P^2 = [0.26 \quad 0.44 \quad 0.24 \quad 0.06]$

$$P^2 = \begin{bmatrix} 0.26 & 0.44 & 0.24 & 0.06 \\ 0.20 & 0.47 & 0.26 & 0.07 \\ 0.16 & 0.46 & 0.27 & 0.11 \\ 0.11 & 0.49 & 0.22 & 0.18 \end{bmatrix}$$

2.2 $V_3 = V_0 \times P^{10} = [0.192 \ 0.457 \ 0.236]$

$$P^{10} = \begin{bmatrix} 0.192 & 0.457 & 0.236 & 0.115 \\ 0.192 & 0.457 & 0.236 & 0.115 \\ 0.192 & 0.457 & 0.236 & 0.115 \\ 0.192 & 0.192 & 0.236 & 0.116 \end{bmatrix}$$

Resposta: 0.115

2.3 In the Initial Case there is a stationary distribution
We can do P^∞ to get the stationary distribution
or we can calculate
$\pi = (\pi_1, \pi_2, \pi_3, \pi_4)$, such that $\pi P = \pi$

eigenvector associated with eigenvalue 1

$(P^T - I)\,\pi^T = 0$

S_1 - $\pi_1 = 0.4\pi_1 + 0.2\pi_2 + 0.1\pi_3 + 0\pi_4$

S_2 - $\pi_2 = 0.4\pi_1 + 0.5\pi_2 + 0.4\pi_3 + 0.5\pi_4$

S_3 - $\pi_3 = 0.2\pi_1 + 0.2\pi_2 + 0.4\pi_3 + 0.1\pi_4$

S_4 $\pi_4 = 0\pi_1 + 0.1\pi_2 + 0.1\pi_3 + 0.4\pi_4$

Because one equation is redundant, we replace one of them with $\pi_1 + \pi_2 + \pi_3 + \pi_4 = 1$ and then we solve the equation $\pi = (0.192, 0.457, 0.236, 0.115)$

In Case 1.2 and 1.3 there are no stationary distribution because the markov chains are not irreductible because when the charger fails or starts to charge slower it can't get to the S_1 state. There is only stationary distributions if the markov chains are irreductible